Discrete
and
Distributed Word Representation

Content

- Discrete Text representation

  N-grams model

- Introduction to Distributed word representation
  - Word2Vec
    - ✓ Skip Gram model
    - ✓ Continuous Bag Of Words model

N-grams

N-gram language model predicts the probability of the next word given (n-1) previous words

n=1 Unigram	n=2 Bigram	n=3 Trigram

Example:
"I love natural language processing”

Unigrams (n=1):
["I", "love", "natural", "language", "processing"]

Bigrams (n=2):
["I love", "love natural", "natural language", "language processing"]

Trigrams (n=3):
["I love natural", "love natural language", "natural language processing"]

Natural language is not a collection of words

The cat sat on the mat
The mat sat on the cat
The sat mat on the cat
Sat the mat on the cat

Doesn’t make a meaningful sentence

Language modeling assigns probabilities to a sequence of words

The cat sat on the mat (High probability score).
The sat mat on the cat (Low probability score).

Language Modeling using Markov's Assumption

Example:
The cat sat on the mat.

First-order Markov's Assumption

This is essentially a **bigram model**.
It assumes the probability of the next word depends **only on the previous one**:

$$P(w_t \mid w_1^{t-1}) \approx P(w_t \mid w_{t-1})$$

- **Example:**
  We have bigram counts from the training corpus.
  - P(cat | the)
  - P(sat | cat)
  - P(on | sat)
  - P(the | on)
  - P(mat | the)

Second-order Markov Assumption

This is a **trigram model**.
It assumes the probability of the next word depends on the **previous two words**:

$$P(w_t \mid w_1^{t-1}) \approx P(w_t|w_{t-2},w_{t-1})$$

From the sentence we can extract trigrams:

- P(sat | the, cat)
- P(on | cat, sat)
- P(the | sat, on)
- P(mat | on, the)

Example:
The cat sat on the mat.

Now the model uses the last **two words** to predict the next one.
This will be more accurate as more words are taken into context than in first order Markov's assumption

k-th order Markov assumption

The probability of the next word depends on the **previous k words**:

$$P(w_t \mid w_1^{t-1}) \approx P(w_t \mid w_{t-k}, w_{t-1})$$

For example, with **k=4** (a 5-gram model), after seeing:

"the cat sat on"

The model uses those four words to predict the fifth word.

Limitations

The animal did not cross the road as it was too {wide/tired}.

Capturing long term dependency depends on the value of k chosen.

We need a language model which will considers all kinds of features of the context which might be useful in predicting a word.

Distributed Text Representation

It's a way of representing **words (or larger text units)** as **dense vectors** in a continuous space, instead of as sparse indicators (like one-hot vectors). These are called word embeddings.

Associate a real-valued d-dimensional distributed feature vector with each word in the vocabulary

Properties of these vectors

- These vectors are d dimensional where $d << |v|$, v being the size of the vocabulary
- **Similar words occur together and have similar context.**
- $V_{\text{rose}} = \begin{bmatrix} 1 \cdot 6 \\ 2 \\ -5 \\ 4 \end{bmatrix}$ $V_{\text{marigold}} = \begin{bmatrix} 1 \cdot 4 \\ 2.1 \\ -5.2 \\ 4.3 \end{bmatrix}$ $V_{\text{cat}} = \begin{bmatrix} 7 \cdot 4 \\ -0.1 \\ -1.1 \\ 0.3 \end{bmatrix}$

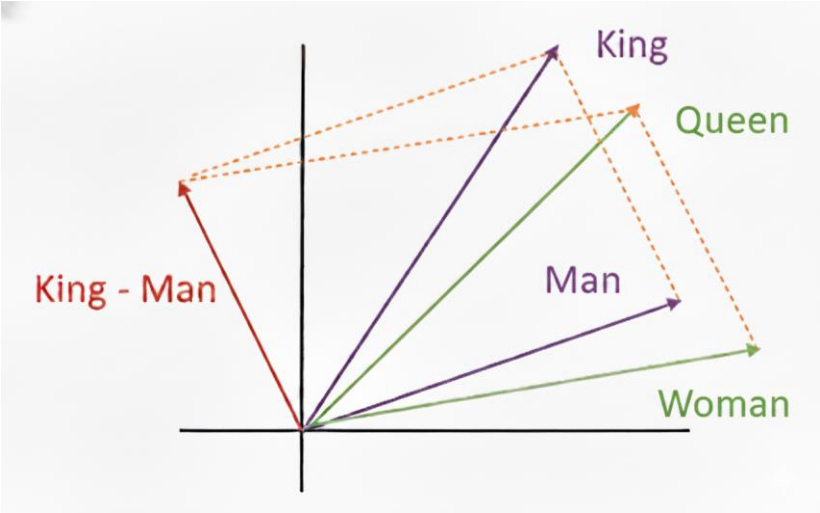
You shall know a word by the company it keeps" – J.R. Firth(1957)

Distributed Text Representation

Core idea:

Similar words tend to occur together and will have similar context

Can the Calculations using these vectors capture semantic association?



$$V_{king} - V_{man} + V_{woman} = V_{Queen}$$

$$V_{king} - V_{man} = V_{Queen} - V_{woman}$$

"royalty" and "human" are preserved
only the gender attribute changes

Word2Vec

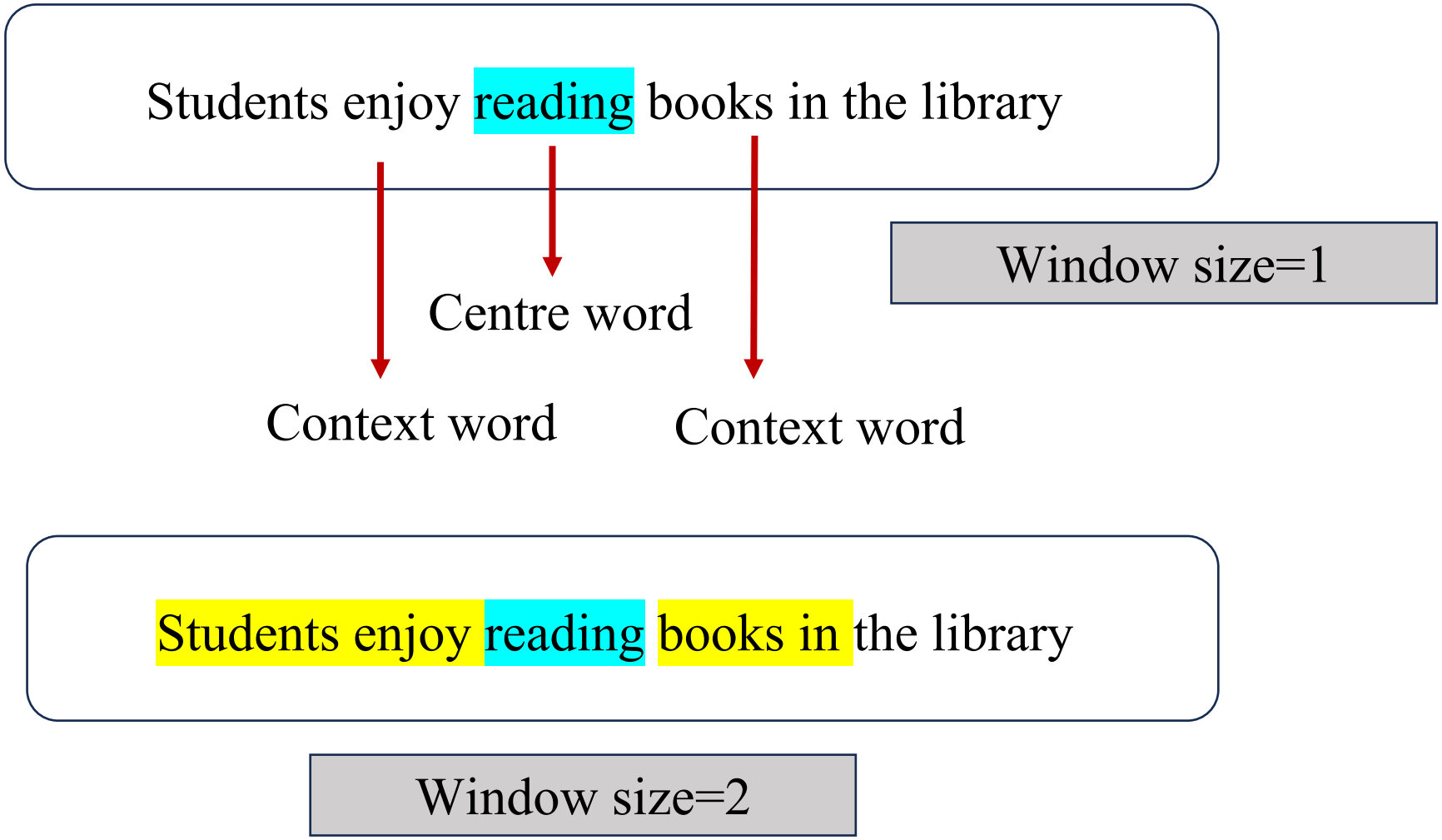
- ✓ It is a technique for converting words into numerical vectors (word embeddings) that capture semantic relationships, meaning words with similar contexts have similar vectors

- ✓ It's a combination of two techniques
  - Skip Gram Model
  - Continuous bag of words.

- ✓ Both of these techniques are shallow neural networks that map words to a target variable which is also a word(s).

- ✓ Both of these techniques learn weights of the neural network which act as word vector representation.

Centre word and Context word in NLP

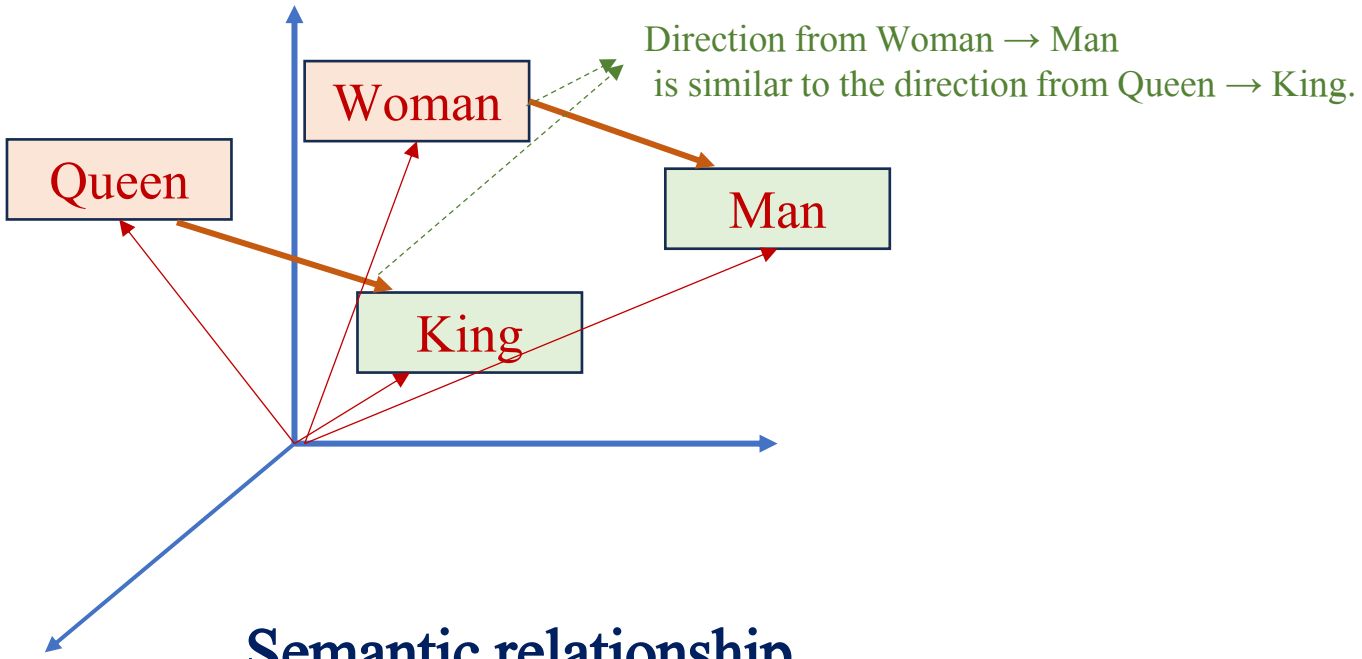


Word2Vec Word Embeddings

Example:

- The **king** and the **queen** rule the kingdom.
- A **man** and a **woman** lead their people

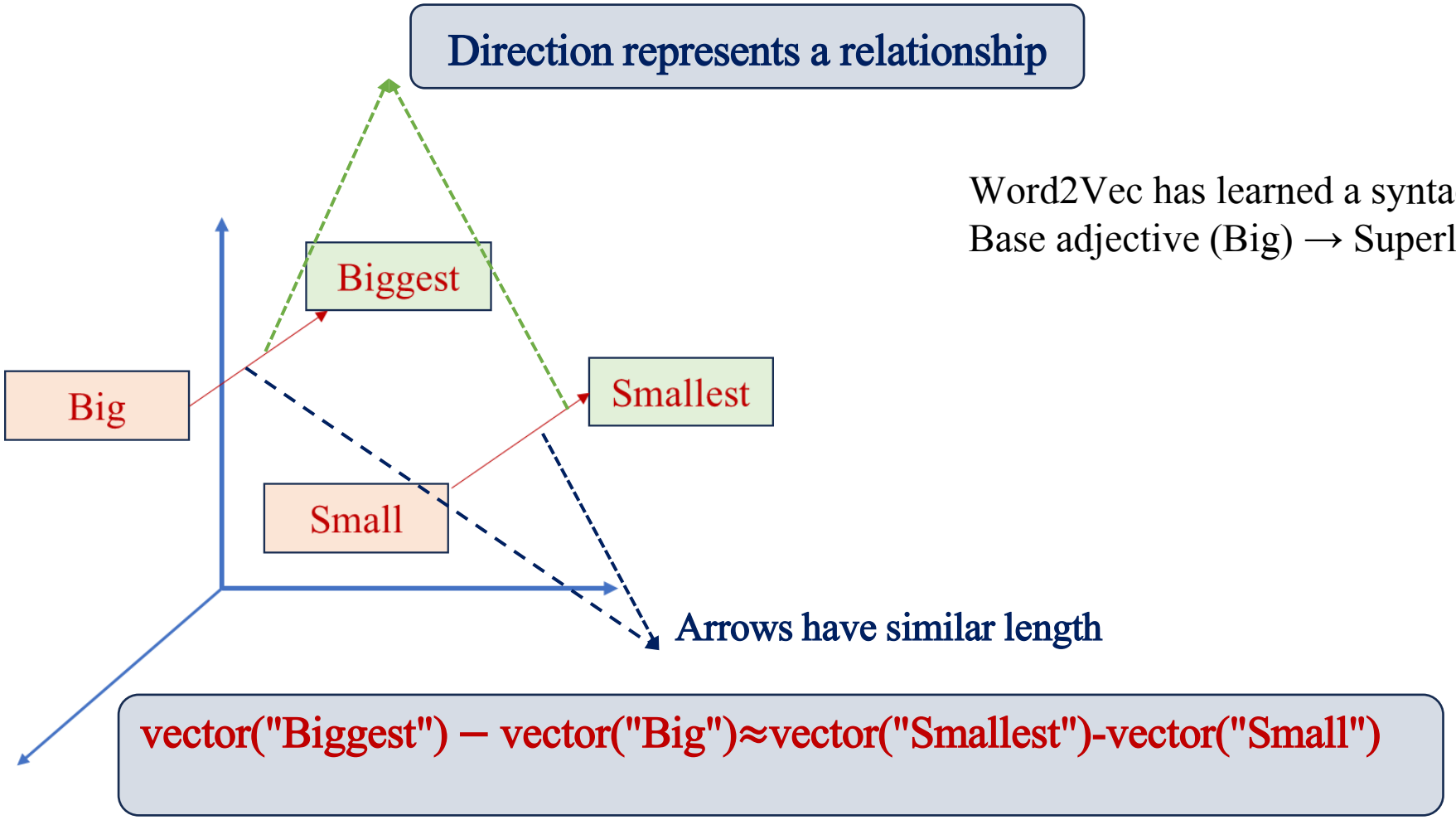
Places them in **similar contexts** (rule, lead, people).



Semantic relationship

Word2Vec Word Embeddings

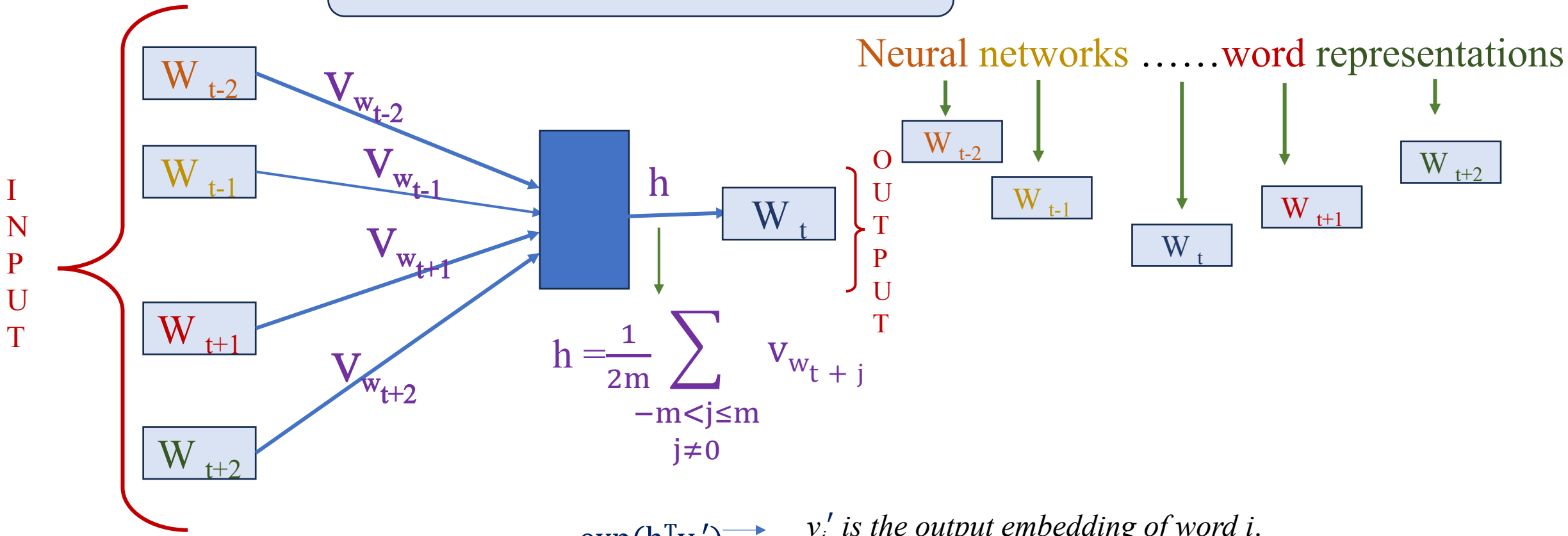
Big, Small, Biggest, Smallest are shown as points in that space



Word2Vec has learned a syntactic relationship:
Base adjective (Big) → Superlative form (Biggest)

Prediction of Centre Word from Context Word(s)

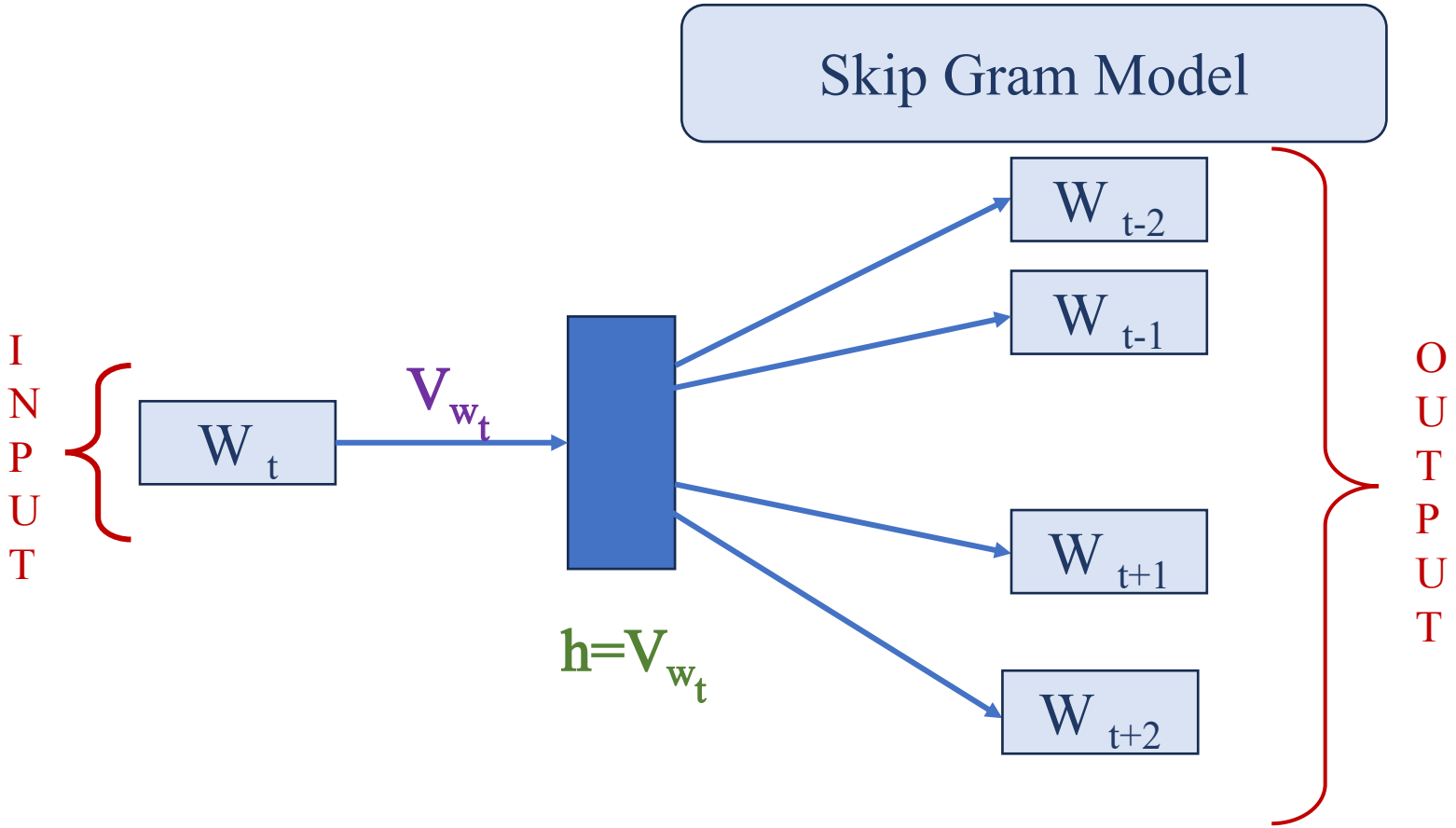
Continuous Bag of Words (CBOW)



$$P(W_t = i \mid \text{Context words}) = \frac{\exp(h^T v_i')}{\sum_{w \in V} \exp(h^T v_w')}$$

v_i' *is the output embedding of word* i.

Prediction of Context Word(s) from Centre Word



$$P(W_{t+j} = i | Wt) = \frac{\exp(h^T v_i')}{\sum_{w \in V} \exp(h^T v_w')}$$

Each word in the vocabulary has an output embedding v'_w.

Summary

Thank you

Next Session

Word embeddings